UGTOMS GSP4 Camera Seed Storage

Camera2 exact evidence, Mali Vulkan compute, UGYUVS1 replay, and bounded non-generative 3D+time

PROFILE 0.3 - PARTIAL-SUBSTRATE CAMERA DRAFT

Target: POCO X7 Pro | 1280 x 720 YUV420 at 30 fps | Mali-G720 MC7
Evidence snapshot: 2026-08-30 | 297-frame POCO capture and Mali dispatch verified

Engineering verdict

The physically implemented path is an asynchronous partial-substrate exact-camera baseline: Camera2 samples are copied into a bounded queue, durably spooled without loss, transformed after capture by an integer Vulkan-compute residual stage, committed as a UGYUVS1 seed/address program plus explicit novelty, and accepted only after native and independent Python replay. It does not yet execute the full Klein/KLB37/tree/kinematic/SDF substrate. The root seed cannot replace arbitrary camera novelty, and metric 3D and hidden surfaces remain UNKNOWN.

WHAT EXISTS AND WHAT DOES NOT

Executive verdict

STATUS	CURRENT EVIDENCE	BOUNDARY
VERIFIED	Portable C++ writer/reader plus independent Python replay	Synthetic exact fixtures
PARTIAL	Seed-regenerated 2D UGLUT2 traversal and owner-only YUV420 codewords	No Klein/KLB37/tree/kinematic/SDF family
PHYSICAL	297 Camera2 frames/results/spooled at 1280 x 720 / 30 fps	10-second POCO run
PHYSICAL	297 Mali integer dispatches; full CPU byte parity	No CPU fallback
VERIFIED	Native on-device replay and independent pulled-file Python replay	All Y/U/V/PTS/metadata
UNKNOWN	Metric 3D, hidden surfaces, complete human body	No promotion from pixels alone

Delivered partial-substrate baseline: Camera2 dense YUV420 is the authority. The seed-regenerated 2D UGLUT2 traversal, GSP4-derived lineage, exact modular residuals, negative-memory omission, durable commits, and two independent replay implementations are concrete. The Mali shader and Vulkan host path execute with byte-for-byte CPU comparison on every dispatch.

Size truth: root seed payload 8 B; standalone capture 177,487,908 B (169.266 MiB); APK 1,582,206 B; file + APK 179,070,114 B. Accepted dense YUV was 410,572,800 B, so this run is 2.31324:1, or 56.7707% smaller.

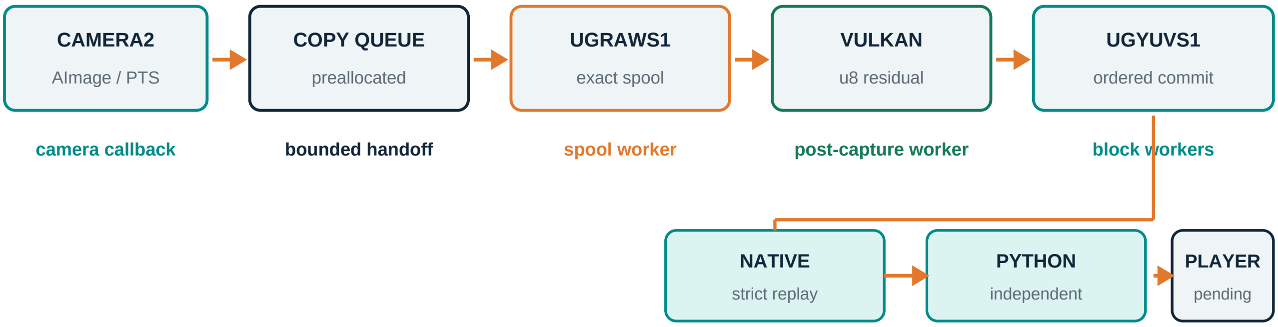
SCOPE CORRECTION: this physical UGYUVS1/Vulkan artifact is not a full-substrate encoding. It does not execute or serialize/hash-bind the discrete Klein quotient, KLB37-style packed log-spherical nodes, parity/radix-tree bifurcation, kinematic phase/velocity/acceleration, periodic lower-case phi cone field, or the pyramid/triangle, sphere, cone and apex SDF operator family.

- No user-supplied MP4 is an input, payload, or acceptance fixture for this camera profile.
- The installed ARM64 APK is 1,582,206 bytes, SHA-256 D421BCD4...2C4841C; its ZIP entries include no .mp4.
- The inspected ARM64 library exposes AImageReader and Vulkan compute symbols but no AMediaCodec or AMediaExtractor symbols.
- The editable KCCH392 pack owns one PLAYER node and one RECORDER node; its current pack is 436 bytes and its UGLUT2 dependency is 144 bytes.
- The POCO accepted 297 Camera2 images with 297 capture results and spooled all 297; the exact spool was 410,672,720 bytes.
- Mali-G720 MC7 executed 297 residual dispatches with full CPU byte parity and no fallback; native replay verified all planes, metadata and sensor PTS.
- The pulled 177,487,908-byte file passed independent Python verification; visible player presentation, host C++ rerun, endurance, and crash-prefix recovery remain open.
- Same-frame exact comparison: UGYUVS1 is 1.7085484454x the FFV1 file and 2.7289156226x the x264 lossless file; it is not codec-competitive yet.

This PDF remains a DRAFT because the exact partial-substrate recorder/transcoder/replay chain is physical, but the full-substrate camera operator, visible PLAYER-node presentation, one- and ten-minute endurance, kill recovery and calibrated 3D promotion are not yet complete.

NO MAIN-THREAD ENCODING

Asynchronous execution contract



UI/main thread: scene lifecycle and controls only - never camera copy, hashing, residual generation, entropy, or file I/O

Camera callback: acquire one AImage, validate dimensions/strides/timestamp, copy dense Y/U/V and the 128-byte canonical metadata record into a reserved slot, release the AImage, and return. It does not hash, encode, submit Vulkan, or write files.

Spool worker: consumes slots in ordinal order, writes the exact UGRAWS1 record, hashes planes and metadata, releases each slot, and fails explicitly on any I/O or sequence error.

Post-capture worker: starts only after the spool is closed and consistent. It rechecks every spool digest, dispatches Vulkan residual work, feeds bounded native block workers, commits results in canonical order, finalizes UGYUVS1, replays the file, then deletes the spool only after exact verification.

Main/UI thread: only owns activity/scene lifecycle and editable controls. Stop/focus-loss can trigger finalization, but transcode and replay continue on a dedicated worker.

Queue pressure never licenses a drop. The legal outcomes are continued exact spooling or an explicit failure receipt and stop.

SEED, RECIPE, NOVELTY, AND TRANSIENT SPOOL ARE DIFFERENT

What is stored

SEED	UGLUT2	HEADER	FRAMES	END	SHA
8 B root	144 B once	512 B	384 B + blocks	192 B	chain

Final .ugsp4c: seed/program state + exact novelty evidence + integrity/commit state

Transient .camera-exact.spool.partial
dense Y/U/V + metadata; removed only after verified replay

Not stored: W x H permutation, MP4 payload, generated pixels
the traversal is regenerated; camera novelty remains explicit

Artifact	Role	Persistence / size
root_seed_u64	Selects deterministic traversal lineage	8 bytes; addressing only
UGLUT2	Literal binary16 log-radius and direction support	144 bytes once; SHA-bound
KCCH392	Editable PLAYER/RECORDER scene-node ownership	436-byte current pack
UGRAWS1 spool	Crash-survivable exact dense Camera2 handoff	Transient; 1,382,736 bytes/frame with fixed metadata/header
UGYUVS1 .ugsp4c	Final seed program plus exact novelty and integrity chain	Content-dependent; never seed-only for arbitrary camera data
JSON receipt	Independent audit and machine-readable replay result	Derivative; not sample authority

At 1280 x 720 YUV420, one authoritative dense frame is exactly **1,382,400 bytes**. The spool adds one 208-byte frame header and one 128-byte metadata record: 1,382,736 bytes/frame, about 41.48 MB/s at 30 fps.

That is approximately 414.8 MB for 10 seconds, 2.49 GB for one minute, and 24.89 GB for ten minutes before final transcode. These are deterministic capacity estimates, not measured sustained phone throughput.

Measured physical run: 410,572,800 accepted YUV bytes became a 177,487,908-byte standalone .ugsp4c (169.266 MiB). With the 1,582,206-byte APK, the combined stored footprint is 179,070,114 bytes. The 8-byte seed is included in program state; it is not the standalone recording size.

A tiny seed is not a tiny lossless camera recording. If the observations are absent from the file, APK, network, or declared external store, they cannot be reconstructed exactly.

THE PARTIAL BASELINE EXECUTES THE ADDRESS PROGRAM

Literal codeword and 2D log-polar traversal

```
UGCODE24-420(x,y) = [Y(y,x), U(floor(y/2),floor(x/2)),
                      V(floor(y/2),floor(x/2))]

store Y at every generated (x,y)
store U,V only when x and y are both even

a_i = UGTRV1(UGLUT2, combine(root_seed, recipe_seed), i)
```

Every 2 x 2 luma group contains four Y bytes plus one U and one V byte. Canonical owner packing therefore preserves the exact six YUV420 bytes without copying chroma four times.

UGTRV1 regenerates every luma address exactly once from doubled pixel centers, binary16 radius/direction lanes, integer Q16/Q30 conversion, radial and angular classification, packed rho/theta keys, SplitMix64 lineage, and a Cartesian final tie-break.

The file stores the root seed, recipe/profile identifiers and dependency hashes. It never stores a 1280 x 720 permutation. The current 720p synthetic oracle generated the traversal and inverted all Y/U/V bytes exactly.

Seed and recipe are executable structure. Novelty bytes are irreducible observed evidence. The distinction is the core information boundary.

Current fixed-profile evidence

Item	Verified value
Dense YUV420 bytes/frame	1,382,400
Owner-packed codeword bytes/frame	1,382,400
UGLUT2	144 bytes; SHA-256 dde65daf...243b49
Python oracle traversal digest	a3be1412...05fa22
Pack/inverse	Every synthetic Y, U and V byte reproduced

CORRECTION - THE PHYSICAL ARTIFACT IS A PARTIAL BASELINE

Full-substrate gap and next acceptance gate

The 297-frame POCO result verifies a useful exact camera/storage path, but it does not satisfy the user's full-substrate requirement. Every missing operator below must become operative in the encoded program, hash-bound, and independently replayable. A decorative LUT, unused receipt field, or CPU-only side calculation does not close this gate.

Required substrate element	Physical UGYUVS1 baseline	Full-substrate acceptance
Discrete Klein quotient	ABSENT	Versioned seam/gluing rule plus finite seam fixtures
Packed log-spherical nodes	Only 2D UGLUT2 address traversal	KLB37-style fixed record; seed-regenerated rho/theta/phi; generated-node hash
Parity/radix-tree bifurcation	ABSENT	Flat breadth-first radix/BST records; deterministic branch/route hash
Kinematic calculus	ABSENT	Bounded fixed-point phase += velocity; velocity += acceleration
Lower-case phi cone field	ABSENT	Periodic phi, frozen units/range/wrap and guarded field classification
Elementary SDF family	ABSENT	T side-view pyramid/delta triangle, sphere, cone and apex operators
Exact camera residual	PRESENT and physically verified	Must remain byte authority after every generated prediction

Implementation gate: freeze one versioned integer/fixed-point operator, including quantization, overflow, seam, branch, wrap and guard behavior. Hash-bind the generated node program and every selected predictor/route to each frame or block.

Port that same operator to Mali Vulkan, native CPU and the independent Python oracle. Require identical generated nodes, branch hashes, SDF/guard classifications, residual bytes, Y/U/V, metadata and PTS across all three implementations.

Re-encode the same 297-frame fixture and count every stored node, tree, SDF, kinematic and guard byte. Compare the final exact file against the 177,487,908-byte partial baseline, FFV1 and x264 lossless. Until that passes, full-substrate size and compression benefit are UNKNOWN.

NOVELTY IS EXACT DIFFERENCE, NOT INVENTION

Applied GSP4 baseline and negative memory

```
finite generated state
  -> radial/angular address support
  -> lane/profile compatibility
  -> finite buffer/timestamp/predictor guards
  -> verified exact modulo-256 difference
  -> seed-derived route and lineage
  -> append-only novelty chain

delta = (observed - predicted) mod 256
observed = (predicted + delta) mod 256
```

A zero residual is negative memory: the generated prior state already yields the accepted byte, so no novelty value is emitted. A nonzero residual remains in the file. Omission never means empty space, disappearance, occlusion, or a hidden surface.

Each default block covers 65,536 luma addresses and chooses the byte-smallest exact representation under a canonical tie order.

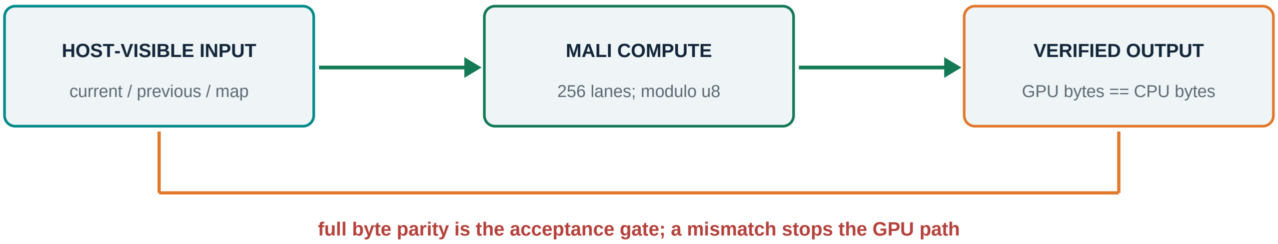
ID	Block representation	Exact payload rule
0	ZERO	No residual values; generated state is already exact
1	DENSE	Every owner-ordered residual byte
2	SPARSE_BITMASK	Occupancy bits followed by nonzero bytes
3	SPARSE_GAPS	Canonical ULEB zero-gaps plus nonzero bytes

The current predictor baseline is RAW_EXACT_LANE for checkpoints and PREVIOUS_SAME_ADDRESS afterward. Spatial MED and temporal-plus-spatial-difference searches remain future exact candidates; adding them must not change sample authority.

Every frame binds ordinal, sensor timestamp, metadata, dependency ordinal, dense-plane hashes, pre-substrate digest, novelty digest, prior-record hash and self hash. Two alternating commit slots expose only a fully committed prefix after interruption.

EXACT INTEGER RESIDUAL, NOT AI

Why Mali Vulkan compute is used



Entropy selection, ordered commits, disk I/O and geometry stay outside this shader.

The shader computes the canonical owner-ordered residual from current dense bytes, prior dense bytes, and a seed-derived lane-source map. It uses 8-bit storage and 256 invocations per workgroup. The host records dispatch count, workgroups, GPU timestamp, wall time and residual SHA-256.

The CPU independently computes the full residual and compares every byte before the writer consumes GPU output. Vulkan failure or mismatch disables the GPU route; it does not silently authorize approximate data.

Physical execution: the POCO selected Mali-G720 MC7 with Vulkan API 1.3.278, 256-lane workgroups and 5,400 groups for 1,382,400 owner-ordered lanes. All 297 frames dispatched on the GPU with no CPU fallback. The first dispatch measured 2.446846 ms GPU and 4.227 ms wall, with full CPU byte parity. Its frame-0 residual SHA-256 is b690a0143df7870c4407c2e9fb33e91526fc4b3cc4978e0e88d17a38610475c6; this is not an aggregate residual hash.

Capability boundary: ADB also reports storageBuffer8BitAccess, shaderInt8, up to 1,024 compute invocations, 32,768 bytes shared memory and 64-byte noncoherent atoms. One run proves this bounded path on this device/software state, not every driver, workload or thermal condition.

Bandwidth: authoritative capture is about 41.47 MB/s. A naive current+previous upload plus residual readback is about 124.4 MB/s, excluding a one-time/static lane map and synchronization. That traffic is modest for an integrated GPU, but host-visible mapping, cache flush/invalidate, readback fences, spool I/O and CPU block packaging can dominate. Physical measurements decide the bottleneck.

NPU: not used. The task is a simple exact integer bijection, while NPU paths add vendor/runtime uncertainty and usually target approximate tensor inference. An NPU may propose non-authoritative features later, never camera-content truth.

EVIDENCE SNAPSHOT - BOUNDED CLAIMS ONLY

Verified partial-baseline device evidence

Check	Result
Python codeword oracle	9 tests pass; strided normalization, owner packing, inverse, residual and lineage
Independent Python UGYUVS1 verifier	7 tests pass; all four block modes, commit recovery, FINAL gates, corruption rejection
Native/Python 320 x 180 cross-fixture	4 exact frames; file SHA-256 22ac87ed...216d1
Native/Python 1280 x 720 cross-fixture	3 exact frames; 4,147,200 authoritative bytes; file SHA-256 63b51ae1...7cdab
720p file size	1,393,123 bytes for checkpoint + unchanged + five-byte sparse change
720p novelty	1,380,600 checkpoint events, 0 unchanged events, 5 sparse events
720p block modes	15 DENSE, 27 ZERO, 3 SPARSE_GAPS
Native worker settings	4 persistent block workers, max 8 in flight, deterministic ordered commit
Installed APK	1,582,206 bytes; SHA-256 D421BCD4...2C4841C; ARM64
Static payload audit	No .mp4 ZIP entry; 144-byte UGLUT2; 436-byte KCCH392
Static native symbol audit	AImageReader + Vulkan compute symbols; no AMediaCodec/AMediaExtractor symbols
Physical Camera2 capture	297 accepted images = 297 results = 297 spooled; 410,672,720-byte exact spool
Physical Mali transcode	297 dispatches; 6 block workers; 12 in flight; no fallback; CPU full parity
Physical native replay	297 frames; Y/U/V + metadata + sensor PTS verified; spool removed afterward
Pulled-file Python replay	PASS; 177,487,908 bytes; SHA-256 9df3c9c6...514669

The synthetic 720p fixture is intentionally favorable: the second frame is identical and the third changes only five owner-lane bytes. Its ratio does not predict natural-camera file size.

The host benchmark measured about 23.4 MiB/s for a checkpoint and about 32.2-32.7 MiB/s for unchanged/sparse append, with 0.418 seconds to replay three frames. These one-run host numbers are diagnostic, not POCO performance.

The physical final file is 43.2293% of 410,572,800 authoritative plane bytes, a 56.7707% reduction for this run. It contains 126,884,266 novelty events, 150 DENSE blocks and 4,305 SPARSE_BITMASK blocks. This is one capture, not a general compression ratio.

A separate host C++ replay, visible PLAYER-node presentation, endurance and kill-recovery evidence remain open; the existing on-device native replay and independent Python replay do not waive those gates.

EXACT BASELINES - CURRENT UGYUVS1 IS NOT CODEC-COMPETITIVE

Same-frame storage comparison

All exact candidates below were decoded to the same 410,572,800-byte planar yuv420p stream. The authoritative replay SHA-256 is 10ac4aaa25d68465921419a74dfc2f4f15c3269ed11fe26e04465cd93a36f84b. The lossy proxy is included only as a size reference and does not pass that hash.

Format / setting	Bytes	Decoded identity	Measured finding
Raw yuv420p replay	410,572,800	Authority	Uncompressed exact baseline
UGYUVS1 .ugsp4c	177,487,908	Exact SHA match	2.31324:1 versus raw
FFV1 level 3 MKV	103,882,280	Exact SHA match	UGYUVS1 is 1.7085484454x larger
libx264 -qp 0 MKV	65,039,720	Exact SHA match	UGYUVS1 is 2.7289156226x larger
H.264 High 8 Mb/s proxy MP4	9,515,400	SHA differs - lossy	7,686,088 bps actual; not an exact peer

The UGYUVS1 file is also 18.6527006747x the lossy proxy, but that ratio is not a lossless comparison. A small proxy cannot satisfy the profile's byte-identical replay contract.

Storage verdict: both verified conventional lossless baselines win decisively on this capture. UGYUVS1 currently earns its engineering value through seed execution, deterministic lineage, canonical novelty records, crash-safe commits, and auditability - not through a file-size win.

The next compression work is bounded: test additional exact predictor and block programs against the same frame ledger, retain only byte-identical decodes, and report the smallest tested result. Do not claim an absolute minimum.

CLASSICAL GEOMETRY ONLY AFTER REPLAY TRUTH

From exact camera time to bounded 3D+time

IMPLEMENTATION STATE: visible GLES player presentation and 3D reconstruction are NOT IMPLEMENTED in the delivered camera artifact. This page defines a bounded, non-generative promotion path; it does not report reconstructed geometry.

Stage	Allowed operation	Authority / failure state
0 - evidence	Exact Y/U/V, sensor time, crop, exposure and camera metadata	OBSERVED; replayable
1 - camera	Measured intrinsics/distortion/shutter; row-aware ray tubes	BOUNDED_SUPPORT or UNKNOWN
2 - proposals	GFTT/KLT, masks, descriptors, flow, motion partitions	PROPOSAL_ONLY
3 - contraction	H/F/E, Sampson, cheirality, parallax, triangulation, reprojection	Surviving bounded branches
4 - scene/object	Static candidate, independent motion, occluder, unresolved, unknown	Every observed footprint classified
5 - materialize	Same-time sparse points, surfels, sparse voxels or open mesh	Derived from promoted support
6 - rasterize	Display or proposal residual	Downstream; cannot certify itself

The contractor is circular in the useful sense: camera, object association, scale gauge, visibility, pose and geometry branches repeatedly narrow one another while preserving every surviving alternative. It must be outward-rounded and finite; convergence to a pretty answer is not evidence.

Static regions matter as much as people or moving objects. A mask complement, low motion, repeated texture or prediction success is not proof of static geometry or free space.

No face, voxel, surfel or sheet spans two source times. Chrono continuity is lineage between same-time states. A mesh at one instant may exist only where that instant has promoted visible support; topology behind the camera remains UNKNOWN.

MoGe2, DA3, DINOv3, SAM3, ViT, SLAM and similar systems may inspire exact equations or supply hash-bound proposals. Learned depth, features, masks, pose or completion never replace the exact residual and never write authoritative geometry.

BODY4D AS BOUNDED VISIBLE STRUCTURE

Human specialization without a generated body

May be represented	Must stay UNKNOWN or PROXY_ONLY
Visible 2D joints and part masks at one time	Hidden anatomy and unobserved backs
Bounded joint-angle alternatives and cyclic constraints	A single learned pose treated as truth
Same-time articulated transforms with evidence intervals	Cross-time skin sheet or closed topology
Visibility, association and owner-handoff branches	Occlusion inferred from missing pixels alone
Visible open surface or explicitly authored proxy	Cloth, hair, skin continuity without support

A human target needs more state than a generic rigid object: joint graph, part association, self-occlusion, articulation limits, cloth/hair discontinuity, contact and identity continuity. Those fields are specialized constraints, not permission to synthesize a complete person.

Body4D-style outputs can enter as bounded candidate joints, visible parts or a removable proposal seed. They must carry model/code/checkpoint/configuration hashes and cannot promote hidden shape.

Negative-memory events for a person include support birth/tightening/retraction, joint-branch split, chart discontinuity, visibility change, owner handoff and checkpoint seal. Equal accepted state emits no event.

Practical outputs are an editable matchmove proxy, visible-joint motion track, partial surface reference or downstream authored animation aid. They are not a recovered complete physical body, biometric identity, medical model or garment-fit truth.

USE CUSTOM STATE WHERE IT EARNS VERIFICATION

File-format and application ROI

Format / application	Evidence-adjusted ROI	Rule
.ugsp4c / UGYUVS1	HIGH for partial seed-replay research; LOW current storage efficiency	Partial-substrate camera evidence; 1.7085484454x FFV1 and 2.7289156226x x264 lossless on this run
.ugsp4c.partial	HIGH for crash recovery	Expose committed prefix only; label incomplete
UGRAWS1 spool	HIGH for capture safety	Transient and large; delete only after verified final replay
Grove JSON + KCCH392	HIGH for editability	Ordinary scene-node ownership; no bootstrap/global camera
JSON/JSONL receipts	HIGHEST for audit/interchange	Hashes, units, time, status, limits; not sample authority
PLY	HIGH once sparse support exists	Promoted points/surfels only; include units/status
glTF 2.0.1 JSON / USDA	MEDIUM for downstream presentation	Derived visible geometry and animation; not evidence authority
Alembic / OpenUSD	DEFER	Use only after real time-slice geometry exists
OpenVDB	DEFER	Use only for bounded supported sparse voxels; UNKNOWN is not empty
APK	HIGH for the device proof	Distribution shell; audit code/assets separately
PDF	HIGH for human review	Never the only machine authority

Highest practical application order

- Exact POCO camera recorder plus pending editable PLAYER-node presentation and deterministic pull-back verification.
- Editable time scrubber and evidence inspector for YUV, metadata, novelty and lineage.
- Calibrated classical matchmove and partial scene/object reconstruction workstation.
- Human visible-joint and open-surface reference tooling with bounded alternatives.
- After-the-fact RTX search for the smallest exact file among explicitly tested seed/predictor/block programs.

ROI here means engineering leverage supported by repository evidence, not demonstrated revenue. General closed 3D reconstruction, biometric/medical use and universal-format ambitions have low current ROI because their authority and validation are missing.

BOUNDED POCO EVIDENCE AND REMAINING GATES

Physical acceptance ledger

Required receipt field	Current state
Installed APK bytes / SHA-256	PASS - 1,582,206 B / D421BCD4...2C4841C; device hash matches host
Camera ID and YUV_420_888 size/rate	PASS - camera 0; 1280 x 720; 30/30 fps; min 33,333,333 ns
Accepted AImages = results = spooled = dispatched = replayed	PASS - 297 = 297 = 297 = 297 = 297
Unique monotonic sensor timestamps and zero silent drops	PASS - strict reader accepted; 9.857788-second PTS span; 30.027 fps
Exact transient spool	PASS - 410,672,720 bytes; removed only after native replay
Mali dispatch	PASS - Mali-G720 MC7; API 1.3.278; 297 dispatches; no CPU fallback
Frame-0 GPU timing / residual SHA	PASS - 2,446,846 gpu ns; 4,227,000 wall ns; b690a014...10475c6
GPU residual equals independent CPU residual for every frame	PASS - full parity; prepared residual consumed
Final .ugsp4c bytes and SHA-256	PASS - 177,487,908 B / 9df3c9c6...514669
On-device native replay Y/U/V/PTS/metadata	PASS - all 297 frames
Pulled-file independent Python receipt	PASS - finalized; no tail; every hash/record/frame accepted
Separate host C++ rerun	PENDING
Visible GLES player presentation	NOT IMPLEMENTED
One-minute and ten-minute PSS/thermal/battery	PENDING ENDURANCE RUNS
Kill-during-capture committed-prefix recovery	PENDING CRASH TEST

This 297-frame run closes one bounded physical camera/transcode/replay test. The full scene matrix still requires separate static/color/checkerboard and textured-motion recordings, plus any optional person test. None relax privacy, geometry or UNKNOWN rules.

Acceptance requires the same authoritative bytes and timestamps on the phone, native host reader and independent Python reader. A visible image alone is not proof; neither is a successful Vulkan dispatch without replay parity.

If Vulkan is slower, unavailable or mismatched, preserve the exact spool and run the CPU path with an explicit receipt. If storage cannot sustain capture, stop explicitly. Never trade exactness for a silent drop.

NO HALLUCINATED COMPLETION

Hard nonclaims and local evidence register

Claim	Required wording
Lossless	Byte-identical declared dense Camera2 YUV420 + sensor time/metadata; not photons
Seed	Regenerates traversal/program state; arbitrary camera novelty stays explicit
Substrate	Physical file is PARTIAL: 2D traversal/lineage/residual only; full Klein/KLB37/tree/kinematic/SDF operator is open
GPU	Exact residual accelerator with CPU parity; not geometry or AI authority
Phone	One 297-frame end-to-end run is proven; visible playback/endurance/crash gates remain open
3D	Exact pixels do not imply depth; unsupported physical space remains UNKNOWN
Time	No primitive spans source times; continuity is lineage
Static	Low motion or mask complement is not certified static support/free space
Human	Visible bounded support/proxy only; no invented full body
Optimality	Smallest among tested exact programs; never absolute/global minimum

Primary local evidence

- spec/UGTOMS_GSP4_SEED_CAMERA_0_1.md
- native/ugtc4d/YUV_SEED_CAPTURE_FORMAT.md
- native/ugtc4d/yuv_seed_capture.hpp and .cpp
- src/ugts_kc3/gsp4_camera_codeword.py and tests/test_gsp4_camera_codeword.py
- src/ugts_kc3/ugyuvs1.py and tests/test_ugyuvs1.py
- src/ugts_kc3/android_template/project/app/src/main/cpp/chrono_seed_capture_session.cpp
- src/ugts_kc3/android_template/project/app/src/main/cpp/chrono_vulkan_residual.cpp
- tmp/ugyuvs1_native_720p_cross_receipt.json and tmp/ugyuvs1_720p_stdout.txt
- tmp/gsp4_seed_camera_android_device_20260830/build-report.json
- tmp/poco_gsp4_physical_20260830/poco_capture.ugsp4c and poco_capture.verify.json
- ../klb_cuda_arch_test_v0.1.0/klb_cuda_arch_test/README.md and packed-format sources
- ../Unified_Geometric_Topological_Substrate_GPU_Native_Package/UGTS_GPU_Native_Addendum_Package/spec/UGTS_GN_1.1.md
- Parent TODO.md - user-authored directions and live acceptance status

Final position: retain the verified partial baseline, implement and cross-verify the missing full-substrate operator before claiming substrate completion, keep capture/spooling asynchronous, and let classical bounded evidence - never generated completion - decide what can become 3D or 4D.